

Macroeconomics B

Introduction and Chapter 13

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Structure of the Exercise Class

We will use a "Jigsaw" format:

Work in small groups

Each group becomes **expert** in one problem

Then you **teach each other**

Finally, we solve a problem **together**

Goal:

Learn by explaining

Understand the intuition, not just the solution

The Jigsaw Method

Step 1: Expert Phase

Work in groups on one assigned problem

Step 2: Exchange Phase

New groups: each person explains their problem

Step 3: Wrap-Up

Solve one problem together

Clarify mistakes and key ideas

Your Role in This Class

This is not a passive class

You are responsible for **explaining** your problem

Your group depends on you

Preparation is important

Key idea:

You learn more by teaching than by listening

Typical Class Structure

Opening and warm-up

Expert group work

Break

Exchange phase

Break

Wrap-up and joint problem

Important:

Timing may change depending on preparation

Flexibility of the Class

The structure adapts to the situation

If many students are prepared:

Move faster to discussion

If fewer students attend:

Work in smaller groups

If preparation is low:

More guided explanation

Expectations

Prepare before class

Participate actively

Ask questions

Help your peers

This is a collaborative learning environment

Today

We will start with the first problem set

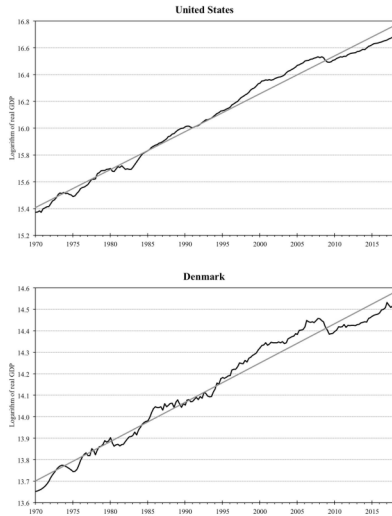
Focus on understanding key concepts

Main takeaway:

The goal is not only to solve problems,
but to understand how to think about them

Bussiness Cycles

Figure 13.1 The evolution of real GDP in the United States and Denmark



Business Cycles

” Business cycles are a type of fluctuations found in the aggregate economic activity of nations that organize their work mainly in business enterprises: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle; this sequence of changes is recurrent, but not periodic; in duration business cycles vary from more than one year to ten or twelve years; they are not divisible into shorter cycles of similar character with amplitudes approximating their own.”

by Arthur Burns and Wesley Mitchell

Measuring Business Cycles

$$HP = \underbrace{\sum_{t=1}^T (y_t - \bar{y}_t)^2}_{\text{cyclical component}} + \lambda \underbrace{\sum_{t=2}^{T-1} [(\bar{y}_{t+1} - \bar{y}_t) - (\bar{y}_t - \bar{y}_{t-1})]^2}_{\text{trend growth component}}$$

Two extreme cases:

- $\lambda = 0$

This implies that $y_t = \bar{y}_t \quad \forall t = 1, 2, \dots, T$.

So all observed fluctuations in y_t reflect changes in the underlying growth trend.

- $\lambda \rightarrow \infty$

This implies that

$$\bar{y}_{t+1} - \bar{y}_t = \bar{y}_t - \bar{y}_{t-1} \quad \forall t = 2, \dots, T-1$$

This means that the growth trend is constant over time.

Note: The HP filter is linear.

Figure 13.3 Cycles in real GDP, six countries, 1970Q1-2015Q3

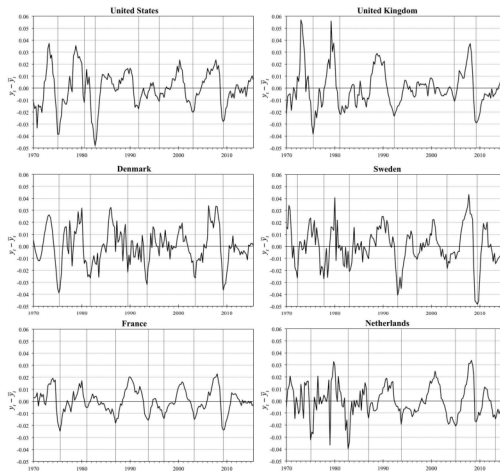
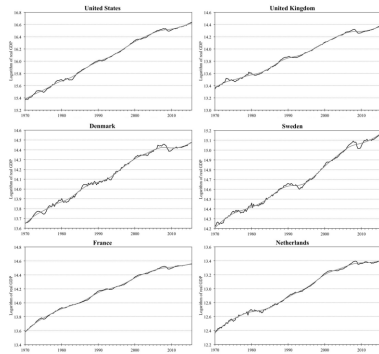


Figure: Output gap (cyclical component)

Measuring business cycles

Figure 13.2 Real GDP and its trend, six countries, 1970Q1-2015Q3



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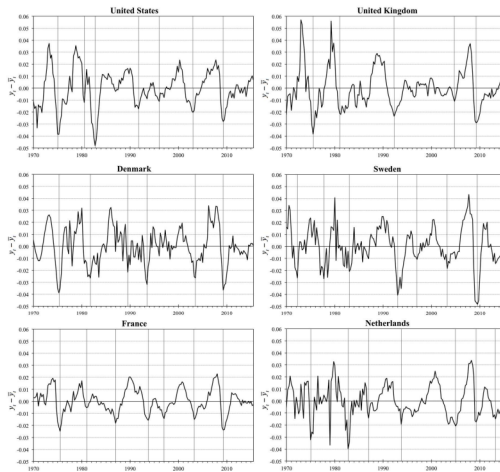


Figure: Output gap (cyclical component)

Identification of the Business cycles

The identification of business cycle troughs and peaks is based on the following simple rules of thumb:

1. A trough must be followed by a peak, and a peak must be followed by a trough.
2. The expansion phase as well as the contraction phase must last for a minimum of two quarters.
3. A business cycle must span a minimum of five quarters.

Volatility

How large is the variability in different economic variables during a 'typical' business cycle?

We may quantify this variability by calculating the standard deviations of the estimated cyclical components of the various time series.

Denmark			
	Average share in GDP	Absolute standard deviation (%) ¹	Relative standard deviation ²
GDP	100%	1.59	1.00
Private consumption	49%	2.06	1.30
Private investment	10%	6.94	4.35
Housing investment	5%	9.58	6.01
Public consumption	26%	1.17	0.73
Public investment	4%	6.42	4.03
Exports	35%	3.15	1.98
Imports	30%	4.50	2.82
Employment		1.08	0.68
Inflation		0.43	0.27

Figure: Output gap (cyclical component)

Volatility

Stylized facts 1 GDP fluctuations are on average more than 1.5 pct. Sometimes 3-4 percent. over and sometimes 4-5 per cent. below.

Stylized facts 2 Investments are very volatile, private consumption is fairly volatile and employment is less volatile.

Stylized Facts 3 The typical deviation for inflation is 0.5 per cent.

Correlation

The correlation coefficient indicates the degree to which x_t and c_t "Moves" together. Calculated by the following formula:

$$\rho_{x,c} = \frac{\text{cov}(x, c)}{\sigma_x \sigma_c}$$

Note $c_t = y_t - y_t$ is the cyclical component.

- If $\rho(x_t, c_t) > 0$ then x_t is pro cyclic which means x_t is moving in the same direction as GDP
- If $\rho(x_t, c_t) < 0$ then x_t counter-cyclic which means x_t is moving in the opposite direction to GDP
- If $|\rho(x_{t-n}, c_t)| > |\rho(x_t, c_t)|$ then x_t is a leading variable A change in x_t n periods before, affects GDP in the current period.
- If $|\rho(x_{t+n}, c_t)| > |\rho(x_t, c_t)|$, then is x_t a lagging variable which means x_t is affected by a fluctuation in GDP, n periods later.

Correlation

Denmark					
	GDP and X_t				
	Quarterly leads and lags				
	-2	-1	0	1	2
X_t (Real variables)					
GDP	0.54	0.74	1.00	0.74	0.54
Private consumption	0.59	0.69	0.73	0.48	0.34
Private investment	0.32	0.47	0.65	0.60	0.59
Housing investment	0.52	0.65	0.69	0.65	0.53
Public consumption	-0.09	-0.02	0.07	0.07	0.06
Public investment	0.22	0.24	0.27	0.27	0.29
Exports	0.10	0.24	0.39	0.33	0.33
Imports	0.45	0.64	0.75	0.65	0.50
Employment	0.26	0.48	0.63	0.70	0.69
X_t (Nominal variables)					
Inflation rate (CPI)	-0.25	-0.16	-0.04	-0.02	-0.02
Short-term nominal interest rate	-0.24	-0.09	0.06	0.23	0.31
Long-term nominal interest rate	-0.31	-0.16	0.00	0.12	0.13

Note: Based on quarterly data from 1971Q1 to 2017Q1. 24 end-point observations excluded. The inflation data, the private investments data and the short-term nominal interest rate data start in 1971Q2, 1972Q1 and 1979Q2, respectively.

Sources: See Table 13.2.

Correlation

Stylized fact 4 Private consumption, investment and imports are strongly positively correlated with GDP.

Stylized fact 5 Employment is strongly correlated with GDP

Stylized fact 6 Inflation displays a weak positive correlation with GDP (see US data)

Stylized fact 7 Employment and nominal interest rates are lagging variables in relation to GDP

Persistence

Persistence indicates the degree of correlation between a variable and its lag. This is calculated by the coefficient of auto correlation:

$$\text{corr}(x_t, x_{t-n})$$

It is said there is a high degree of persistence, if x_{t-n} has displayed high values(in previous periods), there is a greater chance for x_t to display a high value in the (current period)

Persistence

Stylized fact 8 There is significant persistence in GDP and about the same degree of persistence in the private consumption and the private investments

Stylized fact 9 Employment seems to be more persistent than GDP

Denmark				
	Coefficient of autocorrelation			
	1-quarter lag	2-quarter lag	3-quarter lag	4-quarter lag
X_t (Real variables)				
GDP	0.74	0.54	0.32	0.15
Private consumption	0.69	0.53	0.39	0.21
Private investment	0.65	0.51	0.42	0.26
Housing investment	0.85	0.66	0.49	0.34
Public consumption	0.77	0.54	0.36	0.18
Public investment	0.50	0.37	0.16	0.14
Exports	0.69	0.56	0.31	0.13
Imports	0.75	0.48	0.18	-0.05
Employment	0.88	0.73	0.55	0.40
X_t (Nominal variables)				
Inflation rate (CPI)	0.43	0.11	0.03	-0.07
Short-term nominal interest rate	0.71	0.44	0.21	-0.09
Long-term nominal interest rate	0.79	0.45	0.13	-0.14

Figure: Persistence in the danish economy